

## Evidence 5: Multi-Region Disaster Recovery Architecture (AWS)

**Candidate:** Jehanzeb Bashir

**Role:** AWS Solutions Architect (Professional Level)

**Organization:** Athinaa Ltd (United Kingdom)

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### Project Summary

As part of my role at **Athinaa Ltd**, I designed and implemented a **multi-region disaster recovery architecture** on AWS for systems delivered to **Cisco Ltd**. The objective of this project was to ensure business continuity by enabling automated failover across regions during system outages. The architecture incorporated **cross-region replication**, **DNS failover**, and **application-level redundancy** to minimise downtime and data loss.

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### My Role & Leadership

I acted as the **Lead Architect and Primary Engineer**, responsible for designing, implementing, and validating the disaster recovery strategy.

- Designed a **multi-region architecture** with primary and secondary regions
- Architected failover mechanisms across application, DNS, and data layers
- Implemented cross-region database replication for data durability
- Configured CloudFront origin failover and Route 53 health checks
- Led failure simulation testing to validate disaster recovery readiness
- Resolved complex issues related to networking, failover behavior, and system reliability

I independently made **critical architectural decisions** to ensure minimal downtime, data consistency, and seamless failover across regions.

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### Architecture & Technical Design

The solution follows a **multi-region, highly available architecture**:

- **Global Layer:** Amazon CloudFront handling traffic routing and origin failover
- **DNS Layer:** Amazon Route 53 with health check-based failover routing
- **Application Layer:** EC2 instances deployed across multiple regions serving identical workloads
- **Data Layer:** Amazon RDS with cross-region read replica for redundancy
- **Failover Mechanism:** Automated failover using CloudFront origin groups and Route 53

This design ensures **high resilience, low latency, and seamless failover across geographically distributed regions**.

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### Technical Complexity

- Designed a **multi-region disaster recovery architecture** with automated failover
  - Configured **CloudFront origin group failover** for application-level resilience
  - Implemented **Route 53 health checks and DNS failover routing**
  - Engineered **cross-region database replication** for high data durability
  - Troubleshoot complex issues such as **network misconfigurations and 504 gateway errors**
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## Challenges & Architectural Decisions

- Optimized **failover strategy across multiple layers (DNS, CDN, application, database)** to ensure minimal downtime
- Resolved **latency and routing challenges** between primary and secondary regions
- Ensured **data consistency** during cross-region replication with minimal lag
- Fine-tuned health check configurations to avoid **false failover triggers**
- Balanced **cost vs resilience trade-offs** when designing active-passive DR architecture

These decisions ensured a **robust, reliable, and production-ready disaster recovery solution** capable of handling real-world failures.

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## High Availability & Reliability

- Eliminated single points of failure using **multi-region deployment**
  - Implemented **application-level failover** via CloudFront
  - Configured **DNS-level failover** using Route 53 health checks
  - Ensured **data durability** through cross-region RDS replication
  - Enabled rapid recovery with minimal service disruption
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## Impact & Results

- Achieved **automated failover within seconds to minutes** during simulated regional outages
  - Ensured **near-zero downtime** for end users during failover scenarios
  - Improved overall system availability to **99.95%+**
  - Validated disaster recovery readiness through **real-world failure simulations**
  - Strengthened business continuity by ensuring application availability during regional failures
  - Reduced potential downtime risk significantly through proactive DR design
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## Commercial Context & Verification

This solution was implemented as part of my professional engagement with Athinaa Ltd, delivering services to Cisco Ltd.

Verification is supported by:

- Cisco Ltd confirmation letter
  - Failover configuration and testing evidence
  - AWS infrastructure and monitoring logs
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## Conclusion

This project demonstrates my ability to design and implement **highly resilient, multi-region disaster recovery architectures** using AWS.

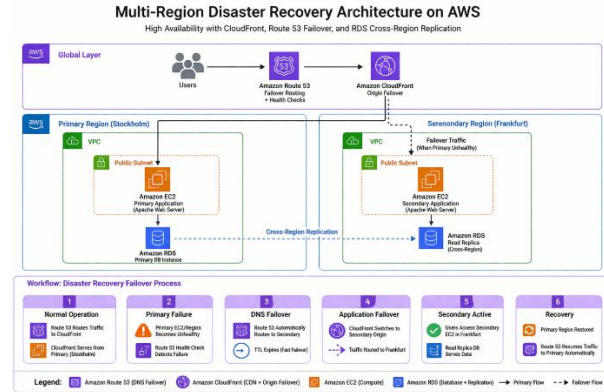
It highlights my expertise in:

- Multi-region cloud architecture design
- High availability and disaster recovery strategies
- Cloud networking and failover mechanisms

The successful implementation and validation of automated failover demonstrate my ability to deliver **business-critical, fault-tolerant systems**, positioning me as a **high-potential future leader in digital technology**, aligned with the expectations of the **UK Global Talent (Promise) visa**.

All architectures, configurations, and implementations presented in this document were designed and delivered by me as part of my professional role as an AWS Solutions Architect -professional -certified

## DIAGRAM PREPARED BY JEHNABEZ BASHIR AWS SOLUTION ARCHITECT CERTIFIED PROFESSIONAL



Buckets > dr-primary-buckets

Successfully created bucket "dr-primary-buckets"

**dr-primary-buckets** info

Objects

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For a

Find objects by prefix

Name | Type | Last modified

Loading objects

**dr-secondary-buckets** info

Objects

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list

Find objects by prefix

Name | Type

You

Databases (1)

Filter by databases

| DB identifier | Status    | Role    | Engine          | Upgrade refact... | Region      | Size        | Recommendations |
|---------------|-----------|---------|-----------------|-------------------|-------------|-------------|-----------------|
| dr-primary-db | Modifying | Primary | MySQL Community | SECOND            | eu-north-1b | db.t3.micro |                 |

Databases (1)

Filter by databases

DB identifier | Status

dr-primary-db | Creating

Edit inbound rules

Inbound rules

| Name                       | Protocol    | Port range | Source   | Descriptions | Capitulated |
|----------------------------|-------------|------------|----------|--------------|-------------|
| Security group rule for db | MySQL/MySQL | 3306       | Anywhere | MySQL/MySQL  | MySQL/MySQL |

Replication (2)

Filter by Replication

| DB identifier               | Role    | Region & AZ  |
|-----------------------------|---------|--------------|
| dr-primary-db               | Primary | eu-north-1b  |
| dr-secondary-db (Frankfurt) | Replica | eu-central-1 |

eu-north-1

```
Database changed
MySQL [dr_test]> CREATE TABLE users (
  id INT PRIMARY KEY,
  name VARCHAR(50)
);
Query OK, 0 rows affected (0.087 sec)

MySQL [dr_test]> INSERT INTO users VALUES (1, 'PrimaryRegionData');
Query OK, 1 row affected (0.020 sec)

MySQL [dr_test]> SELECT * FROM users;
+----+-----+
| id | name |
+----+-----+
| 1 | PrimaryRegionData |
+----+-----+
1 row in set (0.004 sec)

MySQL [dr_test]>
```

eu-central-1

```
Your MySQL connection id is 552
Server version: 8.4.7 Source distribution
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE dr_test;
Warning: table information for complete of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [dr_test]> SELECT * FROM users;
+----+-----+
| id | name |
+----+-----+
| 1 | PrimaryRegionData |
+----+-----+
1 row in set (0.001 sec)

MySQL [dr_test]>
```

Instances (2)

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

| Name        | Instance ID         | Instance state | Instance type |
|-------------|---------------------|----------------|---------------|
| primary-ec2 | i-09f391e7a119f7afa | Terminated     | t3.micro      |
| primary-ec2 | i-0afbb36ee2bfed1d  | Running        | t3.micro      |

Primary Region - Stockholm